Finite-Horizon Phase-Aware Stein Tails for Directional Hardy Fusion

Completing the Schur Packet Route

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Abstract

RH-59 built positive Stein supersolutions in a unitary Schur packet basis. Its all-time observation bound still had a large outer-packet cost. This paper completes that metric with an exact finite-horizon phase calculation. If $\widetilde{O}$ is any positive observability Stein supersolution, we prove the Loewner completion

$$O \preceq O_L + (T^*)^L \widetilde{O} T^L, \qquad O_L = \sum_{m=0}^{L-1} (T^*)^m Y^* Y T^m.$$

The finite term retains all cross-packet phases. For packet tails with Stein upper norms $t_{j,L}$, the fused Hardy energy satisfies the phase-aware bound $\mathcal{E} \leq E_L + \sum_j t_{j,L}$, where E_L is the exact fused finite-horizon response energy. We also give packetwise square-sum completions and geometric tail estimates.

On the five-scale all-column folded-Gaussian audit, the fixed horizon $L = 32$ changes the RH-59 smallest-scale uppers from 19.2196 and 12.1382 to 1.47497 and 1.76413, against exact energies 1.46807 and 1.76031. The finite phase terms are 1.468073 and 1.760309; the remaining tail sums are only 0.006892 and 0.003819. The fitted growth exponents are 0.169 and 0.200, close to the exact-energy exponents 0.168 and 0.199. A 256-bit Arb model certifies the finite-horizon plus tail formula on a two-scalar-block system only. The production result remains binary64 evidence: no uniform physical-family horizon, Stage A1 theorem, arithmetic trace formula, self-adjoint realization, or Hilbert–Polya conclusion is claimed.

Keywords: finite-horizon Gramian; Stein tail; Schur packet; phase coherence; Hardy energy; nonnormal operator; small noise.

MSC 2020: 47A10; 47B65; 93B07; 93D05; 65F35.

1 Introduction

The active small-noise problem in the quadratic transfer program is a pair of directional Hilbert–Schmidt Hardy energies. RH-50 expressed the remaining intrinsic identification gate in this form [5]. RH-56 ruled out uniform fixed-step strong-space contraction [6]; RH-57 showed that fixed radial Riesz blocks become severely oblique [7].

RH-58 moved to a unitary, time-ordered Schur flag and proved dual positive packet Gram identities [8]. That coordinate change removed the oblique projector wall, but a scalar absolute sum over all feed-forward paths grew to 1922.4 and 380.4 at the finest stored level. RH-59 replaced that path ledger by flag-adapted block Lyapunov metrics and exact positive dissipation [4]. The resulting packetwise uppers fell to 19.22 and 12.14, but an outer-packet endpoint cost remained.

The key observation of RH-60 is temporal. The metric is being asked to bound the observation in every direction at every time, including the short times where the actual packet phases have not yet mixed. We can retain the first L time steps exactly and use the positive metric only for the tail. This is not a new choice of Schur weights. It is a completion theorem that can be applied to any valid Stein supersolution.

Contributions and boundary

1. We prove a finite-horizon Loewner completion from an observability Stein supersolution.
2. We prove a global phase-aware Minkowski upper that fuses all finite packet cross terms before summing only the tails.
3. We give packetwise hybrid bounds and a geometric tail corollary.
4. We audit a fixed horizon $L = 32$ on the RH-59 five-scale family and record a horizon sweep through $L = 64$.

The finite-dimensional statements are exact. The production Gramians, Schur forms, inherited metrics, and fitted laws are binary64 diagnostics. The fact that $L = 32$ works on five stored levels is not a continuum or dyadic theorem.

2 Hardy packets and supersolutions

Let A be stable, with source X and observation Y. The directional Hardy energy and observability Gramian are

$$\mathcal{E}^2 = \sum_{m\geq 0} \|YA^m X\|_{\mathfrak{S}_2}^2 = \operatorname{tr}(X^*OX), \qquad O - A^*OA = Y^*Y. \tag{1}$$

Use a unitary Schur form and block partition [1, 2]

$$A = QTQ^*, \qquad T = \begin{pmatrix} D_1 & T_{12} & \cdots & T_{1J} \\ 0 & D_2 & \cdots & T_{2J} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & D_J \end{pmatrix}. \tag{2}$$

Set $\widehat{X} = Q^*X$, $\widehat{Y} = YQ$, and $X_j = E_j\widehat{X}$. The response of packet j is the Hilbert-space vector

$$F_j = (\widehat{Y}T^m X_j)_{m\geq 0} \in \ell^2(\mathfrak{S}_2), \qquad F = \sum_{j=1}^{J} F_j. \tag{3}$$

RH-58's packet Gram is $K_{ij} = \langle F_i, F_j\rangle$, so $\mathcal{E} = \|F\|$ and its diagonal entries are the exact packet energies.

For each packet, upper triangularity leaves the first j Schur blocks invariant. On that prefix let $T^{(j)}$ and $Y^{(j)}$ denote the restricted operator and observation. RH-59 supplies a positive matrix $\widetilde{O}_j$ satisfying

$$\widetilde{O}_j - (T^{(j)})^*\widetilde{O}_j T^{(j)} \succeq (Y^{(j)})^*Y^{(j)}. \tag{4}$$

For its flag metric, this was $\widetilde{O}_j = \kappa_j P_j$; RH-60 does not need that special form.

3 Finite-horizon phase completion

Define the finite-horizon observability Gramian

$$O_{j,L} = \sum_{m=0}^{L-1} ((T^{(j)})^*)^m (Y^{(j)})^* Y^{(j)} (T^{(j)})^m, \qquad O_{j,0} = 0. \tag{5}$$

Theorem 3.1 (Finite-horizon Stein completion)**.** *Assume $T^{(j)}$ is stable and* (4) *holds. Then, for every integer $L \geq 0$,*

$$\boxed{O^{(j)} \preceq O_{j,L} + ((T^{(j)})^*)^L \widetilde{O}_j (T^{(j)})^L.} \tag{6}$$

Consequently, with X_j restricted to the prefix,

$$e_j^2 \leq h_{j,L}^2 + t_{j,L}^2, \tag{7}$$

where

$$h_{j,L}^2 = \operatorname{tr}(X_j^* O_{j,L} X_j), \tag{8}$$

$$t_{j,L}^2 = \operatorname{tr}\Big(X_j^* ((T^{(j)})^*)^L \widetilde{O}_j (T^{(j)})^L X_j\Big). \tag{9}$$

Proof. The exact Gramian splits at time L:

$$O^{(j)} = O_{j,L} + ((T^{(j)})^*)^L O^{(j)} (T^{(j)})^L.$$

Since the tail factor is a positive congruence and $O^{(j)} \preceq \widetilde{O}_j$ by iteration of the positive Stein residual, the second term is bounded by the corresponding congruence of $\widetilde{O}_j$. Testing on X_j gives (7)–(9). □

The finite term is phase-aware because it is not split into packet norms. For $0 \leq m < L$, define

$$F_{j,L} = (\widehat{Y} T^m X_j)_{0 \leq m < L}, \qquad K_{ij}^{(L)} = \langle F_{i,L}, F_{j,L} \rangle. \tag{10}$$

Then $K^{(L)} \succeq 0$ and

$$E_L := \left\| \sum_j F_{j,L} \right\| = \sqrt{\mathbf{1}^* K^{(L)} \mathbf{1}} \tag{11}$$

contains every finite-time cross phase.

Theorem 3.2 (Phase-aware global completion)**.** *Let $t_{j,L}$ be any valid tail upper in* (9)*. Then*

$$\boxed{\mathcal{E} \leq E_L + \sum_{j=1}^{J} t_{j,L}.} \tag{12}$$

Also,

$$\mathcal{E} \leq \sum_{j=1}^{J} \sqrt{h_{j,L}^2 + t_{j,L}^2}. \tag{13}$$

Proof. Decompose $F = F_{<L} + F_{\geq L}$, where $F_{<L} = \sum_j F_{j,L}$. By Minkowski and the triangle inequality,

$$\|F\| \leq \|F_{<L}\| + \sum_j \|F_{j,\geq L}\| \leq E_L + \sum_j t_{j,L}.$$

The packetwise statement follows by applying Theorem 3.1 to each packet and then summing its norm bounds. □

At $L = 0$, (12) is exactly the RH-59 metric absolute upper. As $L \to \infty$ for a fixed finite stable matrix, the tail terms vanish and $E_L \to \mathcal{E}$. The practical question is whether a small, uniform horizon suffices before any continuum limit is taken.

Proposition 3.3 (Normalized metric tail decay). *Suppose $\widetilde{O}_j = \kappa_j P_j$, put $S_j = P_j^{1/2} T^{(j)} P_j^{-1/2}$, and let $W_j = P_j^{1/2} X_j$. Then*

$$t_{j,L} = \sqrt{\kappa_j}\, \left\| S_j^L W_j \right\|_{\mathfrak{S}_2} \le \sqrt{\kappa_j}\, \|S_j\|_2^L\, \|W_j\|_{\mathfrak{S}_2}\,. \tag{14}$$

Proof. Substitute $\widetilde{O}_j = \kappa_j P_j$ into (9) and conjugate by $P_j^{1/2}$. The final inequality is submultiplicativity. □

The first expression in (14), rather than its geometric last expression, is used in the production audit. It retains source alignment inside the normalized flag.

4 Five-scale audit

We inherit the RH-59 all-column folded-Gaussian family, with

$$(\sigma, N) = (0.16, 32), (0.08, 64), (0.04, 128), (0.02, 256), (0.01, 512), \qquad N\sigma = 5.12,$$

Hardy radius $r = 0.85$, and cuts $0.15, 0.35, 0.55$. The RH-59 local metrics, packet scales, and κ_j values are read from its archived binary64 result. No reoptimization is performed in RH-60.

For each $L \in \{0, 1, 2, 4, 8, 16, 32, 64\}$ we compute the exact finite Gram $K^{(L)}$ and E_L, then apply the RH-59 packet tail certificate to obtain $t_{j,L}$. The selected comparison uses the fixed $L = 32$ for all five levels. All finite Grams pass positive-semidefinite binary64 checks; no interval enclosure is claimed.

	exact $\mathcal{E}$		RH-59 metric		RH-60 $L = 32$	
σ	left	right	left	right	left	right
0.16	0.9040	1.0026	1.1719	1.5183	0.9040	1.0026
0.08	1.1626	1.2653	2.1206	2.4307	1.1626	1.2653
0.04	1.3338	1.4845	2.8953	4.0798	1.3338	1.4846
0.02	1.4096	1.6340	3.9855	7.5263	1.4101	1.6349
0.01	1.4681	1.7603	19.2196	12.1382	1.4750	1.7641

Table 1: Exact energies, inherited RH-59 all-time metric uppers, and the phase-aware finite-horizon completion. The last column pair is a valid finite-matrix upper in exact arithmetic; displayed values are binary64.

At the finest scale, the $L = 32$ finite phase terms are 1.4680734 and 1.7603090, while the tail sums are only 0.0068921 and 0.0038190. The selected upper therefore differs from the exact energy by 0.47% and 0.22%. The fitted growth exponents of the selected upper are 0.169 and 0.200, compared with 0.168 and 0.199 for the exact energies and 0.898 and 0.763 for RH-59's all-time metric upper.

L	left phase-aware upper	left tail sum	right phase-aware upper	right tail sum
0	19.2196	19.2196	12.1382	12.1382
4	7.3786	6.0386	6.0341	4.6707
8	3.4136	1.9614	3.0186	1.3570
16	1.7612	0.2934	1.9256	0.1666
32	1.4750	0.0069	1.7641	0.0038
64	1.4681	0.0000	1.7603	0.0000

Table 2: Horizon sweep at $\sigma = 0.01$. At $L = 0$ the completion is the RH-59 metric bound. The $L = 64$ tail values are below 5×10^{-6} in both directions.

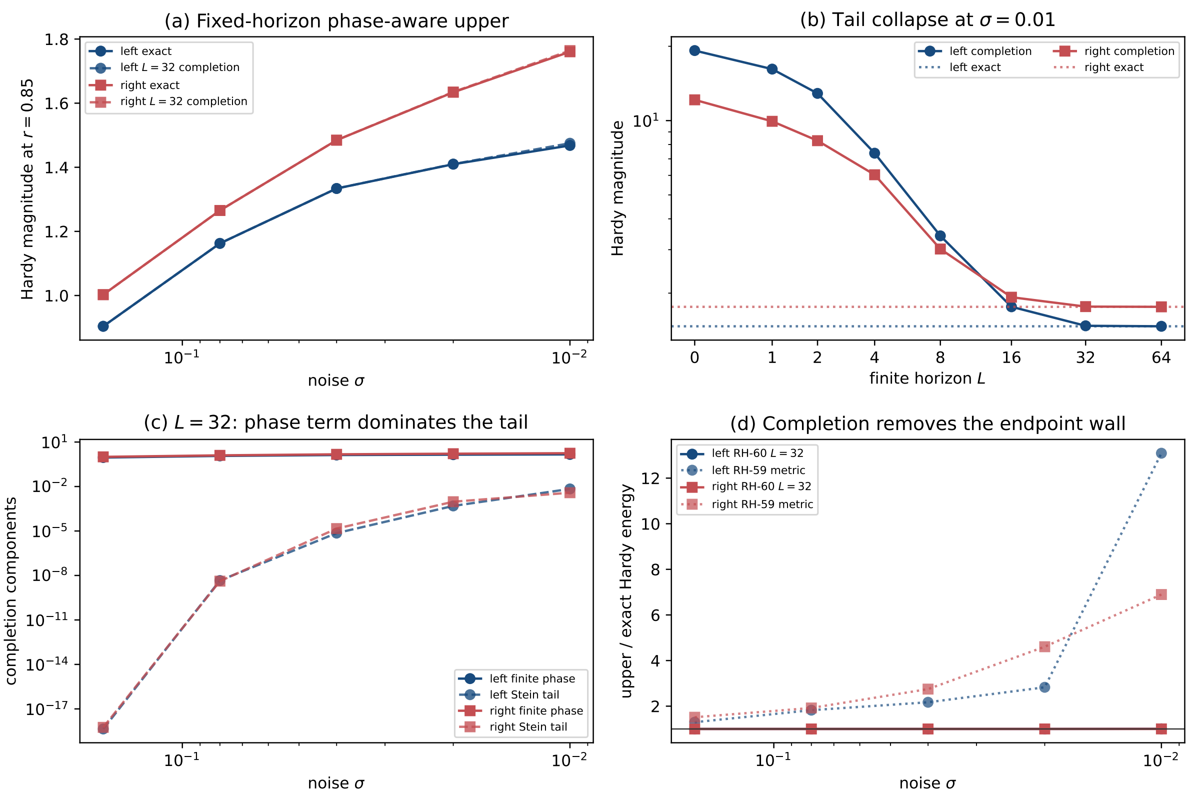

Figure 1: Phase-aware completion audit. (a) Fixed $L = 32$ uppers track the exact energies. (b) The horizon sweep collapses the RH-59 endpoint wall. (c) At $L = 32$ the finite phase term dominates the residual tail. (d) The completion ratio stays near one while the all-time metric ratio grows.

4.1 Outward-rounded model audit

A 256-bit Arb calculation [3] uses the real two-block model

$$T = \begin{pmatrix} 0.2 & 0.3 \\ 0 & 0.7 \end{pmatrix}, \quad Y = (0.8, 0.6), \quad X = (0, 0.4)^T, \quad (s_1, s_2) = (0.25, 1),$$

and horizon $L = 8$. The supersolution multiplier is inflated by $1 + 10^{-6}$ so that its interval residual is strictly positive. Arb certifies the finite energy square, the exact full energy square, the tail upper, and a positive completion gap. The model certificate does not interval-validate a production Schur form.

5 Program consequence

RH-59 all-time metric	Remains a valid finite-dimensional tail engine, but is too pessimistic when applied from time zero.
Finite phase Gram	Restores the observed cross-packet cancellation over a short time window.
Fixed $L = 32$ audit	Nearly closes the stored five-scale numerical budget and matches the exact finite-range growth exponent.
Uniform theorem	Still open. The horizon, Schur packets, and inherited metric weights have not been controlled as a physical continuum family.
Stage A1	Not closed. RH-60 supplies a sharper decomposition of the target, not a dyadically uniform bound.
Stage A4	Still conditional on Stage A1; no later factor or cutoff interface changes.

The next analytic target is now sharply defined: prove a physical-family finite-horizon Gram estimate and a matching tail estimate with L bounded or growing only polylogarithmically in the noise. If that target fails, the failure should be localized to either short-time packet phase control or the long-time outer tail, rather than hidden in an all-time norm.

6 No arithmetic or Hilbert–Polya conclusion

Nothing here constructs a self-adjoint operator, a $T \log T$ counting law, a von Mangoldt or prime-power trace formula, or a zeta-zero identity. No Riemann-hypothesis conclusion is drawn. The independent twin-prime branch is not used.

7 Reproducibility

The archive contains the finite-horizon algebra, tests, horizon sweep, five-scale results, Arb model certificate, figures, dependency hashes, and publication artifacts. Principal commands are

```
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/pytest -q -p no:cacheprovider
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/run_phase_tail_pilot.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/run_arb_phase_tail_audit.py
MPLBACKEND=Agg PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The finite-horizon Loewner completion, phase-aware Gram fusion, packet completion, and geometric tail estimate are analytic finite-dimensional results. Production asymptotics remain numerical. Stage A1, unconditional intrinsic identification, and every arithmetic or Hilbert–Polya conclusion remain outside the claims.

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